

5 Data Analysis and Empirical Results

This chapter presents the empirical findings in four parts. Section 5.1 reports descriptive statistics. Section 5.2 presents the regression results for H1–H3. Section 5.3 reports the event study. Section 5.4 presents the ClimateBERT robustness check. Throughout, coefficients are reported in percentage points (pp), as the dependent variable is the Greenium measured in percentage-point yield differences.

5.1 Descriptive Statistics

Table 1 reports summary statistics for the full panel of 1,455 observations across all four bond pairs. After first-differencing and dropping the first observation within each pair, the working regression sample is 1,451 observations.

Table 1: Descriptive Statistics — Full Panel (Levels, $N = 1,455$)

Variable	Mean	Std	Min	P25	Median	P75	Max
Greenium (pp)	−0.028	0.081	−0.300	−0.073	−0.028	+0.011	+0.276
Sentiment	−0.037	0.078	−0.290	−0.087	−0.036	+0.013	+0.240
Signaling Noise	0.394	0.063	0.179	0.352	0.393	0.434	0.711
VIX	15.978	3.871	11.860	13.190	14.500	17.940	38.570
CDS Spread (bps)	82.319	13.780	52.880	74.230	81.720	84.100	166.880

The mean Greenium across the full panel is −0.028 pp, confirming that sovereign green bonds trade, on average, at a slightly lower yield than their conventional twins. However, the distribution is wide (standard deviation of 0.081 pp) and includes large positive values (maximum of +0.276 pp), indicating that the green bond occasionally trades at a *higher* yield than its counterpart. There is therefore substantial heterogeneity in Greenium dynamics both across and within countries.

Breaking down by country reveals a meaningful difference. For India, the mean Greenium is −0.019 pp ($N = 564$, $SD = 0.042$), while for Indonesia the mean is −0.034 pp ($N = 891$, $SD = 0.098$). Indonesia’s Greenium is both larger in magnitude and substantially more volatile, with a minimum of −0.300 pp and a maximum of +0.276 pp. At the bond-pair level, the India 10Y pair has a mean Greenium of +0.138 pp, reflecting that the green bond trades at a *higher* yield on average—likely because the 10-year green bond (issued January 2023) is priced at a premium relative to an older, off-the-run conventional twin (issued 2019). The India 5Y pair shows a near-zero mean Greenium of −0.000 pp, and the Indonesia 10Y pair shows a conventional negative Greenium of −0.055 pp.

The daily sentiment score averages −0.037 across the panel (India: −0.063; Indonesia:

-0.087), indicating a slight negative tilt in climate-related news coverage. Signaling noise averages 0.394 and is remarkably stable across days (standard deviation of only 0.063), suggesting that internal contradiction in climate media coverage is a persistent structural feature of both countries’ news environments, rather than episodic.

Table 2 presents the pairwise correlation matrix, which provides an initial picture of bivariate relationships prior to controlling for fixed effects and other covariates.

Table 2: Pairwise Correlation Matrix

	Greenium	Sentiment	Sig. Noise	log(VIX)	CDS
Greenium	1.000				
Sentiment	+0.069	1.000			
Sig. Noise	-0.062	-0.092	1.000		
log(VIX)	-0.390	-0.092	+0.034	1.000	
CDS Spread	-0.332	-0.235	-0.097	+0.550	1.000

Two features stand out. First, sentiment has a positive raw correlation with the Greenium ($r = +0.069$), which is the *opposite* sign to what H1 predicts—positive sentiment should widen the Greenium (make it more negative). This is an early warning that H1 may not hold. Second, the VIX carries by far the strongest correlation with the Greenium ($r = -0.390$), establishing that global risk appetite is the dominant driver of day-to-day Greenium variation even before multivariate controls are applied. CDS spreads are also strongly negatively correlated with the Greenium ($r = -0.332$), reflecting the role of country-specific sovereign default risk.

5.2 Regression Results

Table 3 presents the pooled regression results for models M1–M5, estimated on the first-differenced panel of $N = 1,451$ observations with HC3 robust standard errors. All continuous regressors are standardised (mean 0, SD 1) so that coefficients represent the change in the Greenium in percentage points associated with a one-standard-deviation change in each predictor.

Table 3: Pooled Regression Results — First-Differenced Panel ($N = 1,451$)

Variable	M1	M2	M3	M4	M5
$\Delta\text{Sentiment}$ (H1)	+0.0008	+0.0009	+0.0005	+0.0005	+0.0009
$\Delta\text{Signaling Noise}$ (H2)			−0.0031***	−0.0030***	
$\Delta\text{Sent} \times \Delta\text{Noise}$				+0.0001	
$\Delta\log(\text{VIX})$		−0.0034**	−0.0034**	−0.0034**	−0.0035***
$\Delta\text{CDS Spread}$		−0.0021	−0.0021	−0.0021	−0.0024
$\Delta\log(\text{FX Rate})$		+0.0013	+0.0012	+0.0012	+0.0013
Post Treatment (H3)					−0.0298
R^2	0.0004	0.0085	0.0129	0.0129	0.0200
Adj. R^2	−0.0024	0.0037	0.0074	0.0067	−0.0042
N	1,451	1,451	1,451	1,451	1,451

* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$ HC3 robust SE. Standardised regressors. Pair FE: M1–M5. Time FE: M5.

5.2.1 H1: The Credibility Premium (Sentiment Effect)

H1 predicts that positive climate sentiment should widen the Greenium, implying a negative coefficient on $\Delta\text{Sentiment}$. Across all five specifications, the sentiment coefficient is positive, ranging from +0.0005 to +0.0009 pp. None of these estimates is statistically significant at any conventional threshold. The null result is robust to the inclusion of financial controls (M2), signaling noise (M3), the interaction term (M4), and monthly time fixed effects (M5). The magnitude and sign of the sentiment coefficient remain essentially unchanged throughout this progression. **H1 is not supported.**

This finding implies that daily variation in the tone of climate-related news coverage does not translate into measurable changes in the yield premium that investors assign to sovereign green bonds. The result is consistent with the view that sovereign bond markets, unlike equity markets, respond primarily to low-frequency macroeconomic fundamentals rather than to daily textual sentiment signals (Gürkaynak et al., 2005).

5.2.2 H2: The Signaling Noise Penalty

H2 predicts that signaling noise should reduce the Greenium, implying a positive coefficient. The results in M3 and M4 directly contradict this prediction. The signaling noise coefficient is −0.0031 pp in M3 ($p < 0.01$) and −0.0030 pp in M4 ($p < 0.01$), and it survives the addition of the interaction term unchanged. This negative sign means that days with *greater* internal contradiction in climate media coverage are associated with a *wider* Greenium. **H2 is contradicted by the data.**

The interaction term $\Delta\text{Sentiment} \times \Delta\text{Noise}$ in M4 is essentially zero (+0.0001) and

entirely insignificant, confirming that signaling noise does not moderate the relationship between sentiment and the Greenium. The noise effect operates as a direct main effect.

Three interpretations of this unexpected negative sign are plausible. First, days of high media dispersion may coincide with days of heightened climate salience, when competing narratives draw increased investor attention toward green assets and temporarily compress the yield. Second, the dispersion measure may partly proxy for days of elevated news volume, and greater overall media attention toward climate topics—regardless of tone—may favour green bonds. Third, the result may reflect a confound with periods of high global environmental policy activity (such as COP conferences or G20 summits), when simultaneous optimistic and pessimistic coverage is most prevalent and green assets may receive structural support from investor positioning. Disentangling these mechanisms would require richer micro-level data on investor behaviour and article-level attribution.

5.2.3 H3: The Structural Break

H3 predicts that major political shocks produce a negative and significant Post-treatment coefficient, indicating a sustained widening of the Greenium. In the pooled M5 specification, the Post coefficient is -0.0298 pp—correctly signed but statistically insignificant.

H3 is not supported in the pooled specification.

Table 4: Country-Specific DiD Results (M5 Specification)

Variable	Pooled	India Only	Indonesia Only
Post Treatment (H3)	-0.0298	$+0.0209$	-0.0301
Δ Sentiment (H1)	$+0.0009$	$+0.0000$	$+0.0013$
$\Delta \log(\text{VIX})$	-0.0035^{***}	-0.0024	-0.0036^{**}
Δ CDS Spread	-0.0024	$+0.0069$	-0.0026
$\Delta \log(\text{FX})$	$+0.0013$	-0.0053	$+0.0021$
R^2	0.020	0.013	0.028
Adj. R^2	-0.004	-0.029	-0.009
N	1,451	562	889
$^{**}p < 0.05$ $^{***}p < 0.01$ HC3 robust SE. Pair and monthly time FE included.			

The country-specific results in Table 4 reveal a striking divergence. For India, the Post coefficient is $+0.0209$ pp—positive and insignificant, indicating that if anything the Greenium *narrowed* slightly following the auction, contrary to the hypothesis. For Indonesia, the Post coefficient is -0.0301 pp—correctly signed and similar in magnitude to the pooled estimate, but still statistically insignificant.

The most consistent finding across all specifications and both countries is the VIX. In the

pooled M5, $\Delta \log(\text{VIX})$ is -0.0035 pp ($p < 0.001$); in Indonesia only, -0.0036 pp ($p < 0.05$). Even in the India subsample the direction holds (-0.0024), though it falls short of significance given the shorter sample. The interpretation is that a one-standard-deviation increase in $\log \text{VIX}$ is associated with a widening of the Greenium by approximately 0.0035 pp—investors appear to rotate into green bonds during periods of global volatility, compressing green yields relative to conventional ones. This finding echoes Dragotto et al. (2025), who document that macro-financial conditions dominate green bond pricing dynamics at high frequencies.

5.2.4 Model Fit and Diagnostics

Overall model fit is low throughout. R^2 ranges from 0.0004 in M1 to 0.0200 in M5, and adjusted R^2 is negative in M1 (-0.0024) and M5 (-0.0042), reflecting that these models explain very little of the day-to-day variation in the first-differenced Greenium. This is consistent with the view that Greenium movements are dominated by idiosyncratic daily noise and structural liquidity differences between paired bonds, rather than by the political and macroeconomic variables included in the model.

The Durbin-Watson statistics confirm the absence of residual serial correlation: $DW = 2.543$ for M3 and $DW = 2.572$ for M5. All Variance Inflation Factors are below 1.05 , confirming the absence of multicollinearity. The Jarque-Bera test rejects normality of residuals in both specifications ($p < 0.001$), which is expected for daily financial data; the use of HC3 robust standard errors ensures valid inference regardless of the residual distribution (MacKinnon & White, 1985).

5.3 Event Study Results

Table 5 summarises the event study findings across the three samples.

Table 5: Event Study Summary (± 10 Weeks Around Treatment)

	Pooled	India	Indonesia
N (observations)	149	55	94
Mean pre-event coef (pp)	-0.040	0.000	-0.040
Significant pre-event weeks	$0 / 9$	$0 / 9$	$0 / 9$
Mean post-event coef (pp)	-0.055	-0.001	-0.052
Significant post-event weeks	$0 / 10$	$0 / 10$	$1 / 10$
Parallel trends	✓Supported	✓By construction	✓Supported
Treatment effect (H3)	× Not detected	× Not detected	~ Weak (Week 2 only)

5.3.1 Parallel Trends Validation

In the pooled and Indonesia-only specifications, all nine pre-event week coefficients (weeks -10 to -2 , with week -1 as the reference) are statistically insignificant. The mean pre-event coefficient is -0.040 pp in both the pooled and Indonesia samples, with confidence intervals that comfortably span zero throughout. There is no evidence of a pre-existing trend in the Greenium that would confound the post-event comparison. The parallel trends assumption is therefore empirically supported for Indonesia, and holds by construction for India through the twin-bond matching design.

5.3.2 Treatment Effect: Pooled and India

In the pooled specification ($N = 149$), all ten post-event week coefficients are statistically insignificant. Point estimates range from -0.076 pp in the treatment week itself (week 0) to -0.043 pp in week 6, without any systematic trend. The 95% confidence intervals for all post-event weeks span zero. The India-only event study ($N = 55$) similarly shows no significant post-event coefficients, with point estimates oscillating in a narrow band around zero (mean: -0.001 pp). The India inaugural auction produced no detectable sustained change in the Greenium at weekly frequency.

5.3.3 Treatment Effect: Indonesia

The Indonesia event study ($N = 94$) is the only specification that yields any evidence of a treatment effect. Week 2—two weeks following the JETP announcement—produces a coefficient of -0.094 pp ($p < 0.05$, 95% CI: $[-0.201, +0.013]$), indicating a transient widening of the Greenium. No other post-event week achieves statistical significance. The mean post-event coefficient is -0.052 pp, suggesting a modest downward shift across weeks 0–10, but the evidence rests on a single marginally significant estimate whose confidence interval nearly includes zero.

This finding is best interpreted as weak, suggestive evidence that the JETP announcement may have briefly improved investor sentiment toward Indonesian sovereign green assets, consistent with the “Global Validation Shock” characterisation discussed in the Methodology. However, the effect dissipated rapidly, which is consistent with the structural contradictions in Indonesia’s energy policy documented by Neumann (2025): the simultaneous expansion of domestic coal capacity through captive power plant approvals would have reasserted signaling noise in the weeks following the G20, limiting any durable Credibility Premium from the international announcement.

5.4 ClimateBERT Robustness Check

The ClimateBERT robustness results are reported for completeness but must be interpreted with extreme caution. Following the two-stage ClimateBERT pipeline applied to GDELT DOC API headlines, only 54 climate-relevant country-day scores were produced from 1,539 English-language articles. After first-differencing, the regression subsample contains only $N = 20$ observations (Indonesia: 19, India: 1), far below the threshold for reliable inference.

In M3b (ClimateBERT replacing V2Tone), the ClimateBERT coefficient is +0.0145 pp and signaling noise is +0.0046 pp; neither is significant. In M3c (both measures simultaneously), the ClimateBERT coefficient is +0.0210 pp and V2Tone is +0.0109 pp, again both insignificant. The R^2 values appear inflated (0.260 and 0.285), reflecting severe overfitting in a sample where the number of parameters approaches the number of observations.

The ClimateBERT results therefore neither confirm nor contradict the main V2Tone analysis. The sparse coverage is itself the primary finding: the GDELT DOC API is insufficient for producing reliable daily ClimateBERT scores in an emerging-market context, and any future replication using domain-specific NLP would require either more powerful article retrieval infrastructure or a multilingual ClimateBERT variant capable of processing Hindi- and Indonesian-language sources (Webersinke et al., 2022).

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